AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2018

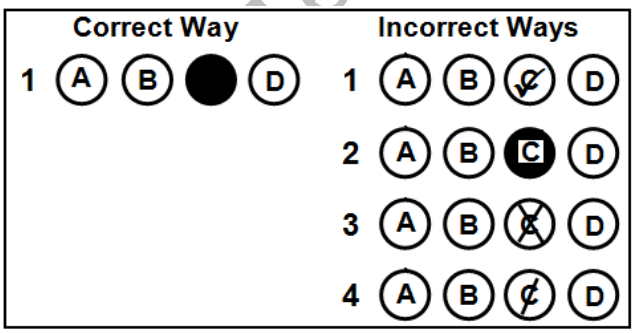
Chemistry Paper I

Time: 35 minutes Marks: 25

AKU-EB May 2018 for Teaching & Learning Only

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. What is the valency of iron (Fe) in $Fe(NO_3)_3$?

   A. 0
   B. 1
   C. 2
   D. 3

2. $\underline{\mathbf{w}}\,NH_{3(g)} + \underline{\mathbf{x}}\,O_{2(g)} \rightarrow \underline{\mathbf{y}}\,NO_{(g)} + 6H_2O_{(g)}$

   The values of **w**, **x** and **y** in the given chemical equation are

	w	**x**	**y**
A	2	5	2
B	2	3	2
C	4	5	4
D	4	3	4

3. A compound **X** has an empirical formula C_4H_9 and its molecular mass is 114 g/mol.

   Which of the following represents the molecular formula of compound **X**?

   (**Note**: ${}^{12}_{6}C$ and ${}^{1}_{1}H$)

   A. C_6H_{11}
   B. C_7H_{12}
   C. C_8H_{18}
   D. $C_{12}H_{27}$

4. Which of the following options shows the sub-atomic particles of a species having a +1 charge on it?

	Number of Electrons	**Number of Protons**	**Number of Neutrons**
A	18	20	20
B	18	19	20
C	10	9	10
D	10	8	8

5. Rutherford's atomic model explains all of the following statements EXCEPT that the

   A. volume occupied by the nucleus is very small.
   B. positively charged particles exist inside the nucleus.
   C. entire mass of the atom is concentrated in the nucleus.
   D. energy emitted by the electrons forms a line spectrum.

BKU-EB May 2018 for Teaching & Learning Only

6. Goiter, a disease affecting thyroid glands, can be diagnosed by the use of

   A. cobalt-60.
   B. iodine-131.
   C. strontium-90.
   D. uranium-235.

7. In a molecule of ethyne (C_2H_2), how many electron pairs are involved in the bonding between carbon-carbon atoms?

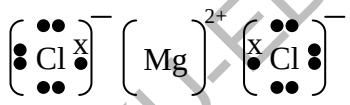
   A. 2
   B. 3
   C. 4
   D. 5

8. Read the given characteristics.

   - Usually dissolve in water
   - Carry partial charges on atoms
   - Uneven distribution of electrons

   Based on the given characteristics, an example of polar covalent molecule is

   A. O_2
   B. Cl_2
   C. H_2S
   D. NaCl

9. 

   (2,8,8) (2,8) (2,8,8)

   In the given dot and cross structure of magnesium chloride, magnesium ion is bonded to two chloride ions by

   A. ionic bonds.
   B. metallic bonds.
   C. covalent bonds.
   D. coordinate covalent bonds.

10. Which of the following statements is FALSE regarding ionisation energy of elements?

    A. The first ionisation energy of lithium is less than that of beryllium.
    B. The second ionisation energy of magnesium is less than that of sodium.
    C. It is the amount of energy required to remove an electron from a gaseous atom or ion.
    D. It is the tendency of an atom or molecule to attract electrons of other atom or molecule to itself.

AKU-EB May 2018 for Teaching & Learning Only

11. Which of the following statements is CORRECT about groups in the periodic table?

    A. Both metals and non-metals are present in all groups.
    B. Reactivity in non-metals increases when moving up a group.
    C. Ionisation energy of elements increases when moving down a group.
    D. The total number of electrons for each element in a group is the same.

12. In which period of the periodic table, an element with atomic number 14 is placed?

    A. 1
    B. 2
    C. 3
    D. 4

13. At constant temperature and pressure, which of the following gas molecules will diffuse the fastest through air?

    (**Note**: Atomic mass of O = 16 amu, Cl = 35.5 amu, N = 14 amu, C = 12 amu and H = 1 amu)

    A. $O_{2(g)}$
    B. $Cl_{2(g)}$
    C. $NO_{2(g)}$
    D. $CH_{4(g)}$

14. At constant temperature, if the volume of a gas in a container is reduced to half, then the pressure on the gas will

    A. be doubled.
    B. reduce to half.
    C. remain the same.
    D. reduce to a quarter.

15. Which of the following substances does NOT have a fixed volume and can be compressed?

    A. Sugar
    B. Honey
    C. Kerosene oil
    D. Water vapour

16. What is the molarity of a solution containing 32 g of potassium sulphate (K_2SO_4) in 350 cm^3 of solution?

    (**Note**: Molar mass of K_2SO_4 = 174 g mol^{-1})

    A. 0.06 M
    B. 0.09 M
    C. 0.52 M
    D. 64.37 M

AKU-EB May 2018 for Teaching & Learning Only

17. The given diagram shows an area covered in fog.



Fog is a solution of

A. gas in liquid.
B. liquid in gas.
C. solid in liquid.
D. liquid in liquid.

18. Which of the following differences between solution and suspension is INCORRECT?

	Solution	**Suspension**
A	It passes through the filter paper.	It leaves residue on filtration.
B	It forms homogeneous mixture.	It forms heterogeneous mixture.
C	It does not show Tyndall effect.	It may show Tyndall effect.
D	It appears translucent.	It appears transparent.

19. Which of the following CORRECTLY matches the redox reaction with its observation?

	Redox Reaction	**Observation**
A	$2HI_{(g)} + Cl_{2(g)} \rightarrow I_{2(g)} + 2HCl_{(g)}$	Hydrogen iodide is reduced to iodine.
B	$Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$	Magnesium is reduced to magnesium chloride.
C	$CuSO_{4(aq)} + Mg_{(s)} \rightarrow MgSO_{4(aq)} + Cu_{(s)}$	Copper sulphate is reduced to copper.
D	$H_2S_{(g)} + Cl_{2(g)} \rightarrow S_{(s)} + 2HCl_{(g)}$	Hydrogen sulphide is reduced to sulphur.

20. The gas(es) produced during the electrolysis of fused sodium chloride in Down's cell is/ are

I. $Cl_{2(g)}$
II. $O_{2(g)}$
III. $H_{2(g)}$

A. I only
B. III only
C. I and II
D. I and III

ACU-EBUG 2018 for Teaching & Learning Only

21. Which of the following is an example of electrolyte?

    A. Benzene
    B. Sulphuric acid
    C. Distilled water
    D. Carbon tetrachloride

22. Metals are good conductors of electricity due to the presence of

    A. freely moving electrons.
    B. number of positive charges on the ions.
    C. atoms in the form of rows one above the other.
    D. electrostatic forces of attraction between oppositely charged ions.

23. Barium belongs to group IIA while chlorine belongs to group VIIA of the modern periodic table.

    If barium reacts with chlorine, then their product is

    A. BaCl
    B. $BaCl_2$
    C. Ba_2Cl
    D. Ba_2Cl_2

24. Compared to a sodium metal, an iron metal

    A. is soft in appearance.
    B. has low melting point.
    C. is an alkaline earth metal.
    D. has stronger metallic bond.

25. Sodium metal is always found in nature in the form of oxides or halides because sodium

    A. is highly electropositive.
    B. is highly electronegative.
    C. has high electron affinity.
    D. has high ionisation energy.

AKU-EB May 2018 for Teaching & Learning Only

END OF PAPER

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only

Please use this page for rough work

AKU-EB May 2018
for
Teaching & Learning Only